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Social Implications of Information Processing

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May 7 2025

The Right to Repair

Cracked your iPhone screen? That will cost you over \$300 to repair at the Apple Store. So why don't people repair their tech by themselves to save money? In many cases, they can't—because many large tech companies implement harmful designs into their devices to limit the consumer's ability to repair them. In many cases, parts will use proprietary screws, be paired together, or even be glued; in combination with sparse product documentation, people are often forced to spend big to fix their devices or even purchase an entirely new device. Each year, millions of devices are scrapped because of companies' design decisions. This started the conversation on the consumer's right to repair their products. The right to repair is the ability to fix the products you own without any purposeful limitations. Limiting consumers' repair ability causes widespread harm to consumers, small businesses, and the environment. By protecting the right to repair, the net outcome results in happier consumers and a safer planet. Corporations need to be held accountable by both their governments and consumers.

To fully understand the importance of the right to repair movement, it is important to know what it truly means, who is impacted, and how it is enforced. Simply described, the right to repair is the idea that any consumer should have full reign to repair the products they own. Implications of restricted repair become much more widespread

than inconvenience—consumers are being extorted to spend far more than they need to. When fewer people repair their products, e-waste, which is detrimental to the environment, increases, small repair shops are going out of business, and much more. Repair restrictions are purposeful and harmful. This issue affects everybody. Some common forms of tech that are being restricted include smartphones, cars, tractors, and medical devices.

Consumers forgo the brunt of the direct impacts of limited repair. Independent repair shops offer a large relief to consumers; in many cases, independent repair shops cost under half as much as manufacturer-licensed repairs [1]. Sadly, manufacturers create physical and legal limitations to stop independent repair shops; this keeps prices high and money in their pockets. A jarring example of this price gouging can be observed with the iPhone. In cases of older phones, such as the iPhone 6, a simple repair such as the screen, when done through Apple, will cost \$115 for a phone worth \$95 [1]. Price isn't the only reason people are forced to buy new tech. Due to the industry's rapid growth, tech can become obsolete shockingly fast. As consumers are constantly forced to cycle through new tech products, they lose their data and sense of ownership. When viewed through the lens of act utilitarianism, if you quantify the loss of convenience and money from consumers, it significantly outweighs the gain from the benefiting corporations. The problems do not stop at the consumer. From here, they trickle into the economy and environment, causing widespread harm.

Restricted repair rakes in profits for manufacturers due to the forced excessive new tech consumption, but they claim that they are working for the best interest of the consumer. A common claim is for the safety of the consumer. For example, any product

with a lithium battery or combustion engine can cause serious issues if repaired incorrectly [8]. Additionally, studies show one-third of dealer repair shops notice illegal modifications to vehicles, such as emission violations [8], proving that open repair is often abused. While these concerns are valid, they are often exaggerated. When repairs are more expensive, these products can often go unmaintained, which can be as harmful, if not more, than self-repair. When viewed from an act utilitarian lens, the risk is even on both sides, but when repair is restricted, the consumer loses large sums of time and money; therefore, these restrictions should not be seen as ethical.

The ethical framework act utilitarianism attempts to model the overall good done to affected parties in order to assess a net positive outcome. In the case of the right to repair, the relevant parties include consumers, repair technicians, and manufacturers. From the perspective of the consumer, both restricted and unrestricted repair can be dangerous in extreme cases. However, when repair is restricted, consumers are forced to spend more money and wait longer for important fixes. Repair technicians benefit from unrestricted repair, as they do not need to worry about punishment or licensing when operating their businesses. Conversely, manufacturers often see greater profit when they restrict repair. Ergo, act utilitarianism would regard restricted repair as unethical. Another framework that offers a more direct decision on repair is Kantianism. Restricting repair for any reason treats the consumer as a means to an end, directly contradicting Kantian ethical principles. Thus, Kantianism also finds restricted repair unethical.

Despite ethical objections, repair remains commonly restricted, which has given rise to the Right to Repair movement. This movement seeks to empower consumers by

encouraging informed decision-making prior to the purchase of new devices and by providing access to otherwise difficult-to-find repair documentation. Prominent tools supporting this effort include iFixit and repair.org. While these resources are valuable, it remains essential for consumers to conduct thorough research before making purchases to avoid potential limitations in future repairability. One key challenge is that even with greater awareness, consumers may face contracts or end-user license agreements (EULAs) that restrict their rights to modify or repair products. These agreements can limit software access, replacement part purchases, or diagnostic tools, further complicating efforts to repair independently.

In response to these challenges, we developed a web application titled the Repairability Calculator. This tool enables users to browse a comprehensive list of products and access information on general maintenance tips, manufacturer warranties, repairability, available documentation, and common hardware faults. The application sources its product list from iFixit's API and utilizes Google's Gemini large language model (LLM) as an alternative to more resource-intensive web scraping techniques. My contributions to the project included identifying viable sources for repairability data, designing and implementing parts of the web application's user interface, and integrating the Gemini model. I selected Gemini primarily due to the impracticality of building a robust scraping algorithm within the project's time frame—particularly since much repair data is not publicly accessible. Instead, Gemini's advanced retrieval capabilities and extensive knowledge base allowed us to efficiently extract relevant product information. The model was prompted with specific queries such as: "Common Issues: List common issues with the product. Warranty: What is the warranty period for

the product, and what terms, voids, or coverage limitations apply? Repair Sources: List repair sources for the product, including links to third-party and self-repair documentation,” along with several format parameters, to standardize output and ensure accuracy. To ensure clarity, we tested multiple prompt variations and compared results across different devices, refining our queries to balance completeness with conciseness.

The right to repair movement continues to grow as consumers, independent technicians, and policymakers push back against restrictive practices. Through our project, we aimed to support this movement by creating a tool that makes repair information more accessible and transparent. While our application provides valuable insights into product repairability, it also highlights ongoing challenges in obtaining accurate, comprehensive repair data. Looking forward, future development could focus on expanding the database, integrating user-contributed repair experiences, and refining the accuracy of model-generated outputs. Additionally, outreach partnerships with advocacy organizations could improve visibility and user adoption. By continuing to develop tools that empower consumers, we can help foster a more sustainable, equitable, and repair-friendly technology ecosystem.

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